Table 18-1 Sensory Receptors: Anatomy, Location, and Function

Receptor	Type	Sensory Nerve Classification	Location	Diameter (μm)	Conduction Velocity (m/sec)	Sensory Function
Sensory (afferent) myelinated	A	Ia	Muscle	12-20	72-120	Muscle spindle primary ending (or 1-degree ending)
	A	Ib	Tendon	12-20	72-120	Golgi tendon organ Rate of muscle shortening
	A	II	Muscle	6-12	36-72	Muscle spindle, secondary ending (or 2-degree ending) muscle length changes
	A	II	Skin	6-12	36-72	Vibration, discriminatory touch, pacinian corpuscles
	A	III	Skin	1-6	6-36	Pricking, sharp pain, temperature (cold), light touch
Unmyelinated	C	IV	Muscle	~1	0.5-2.0	Aching pain
	C	IV	Skin	~1	0.5-2.0	Pain, temperature

From Guyton AC: Sensory receptors and their basic mechanisms of action. In *Textbook of medical physiology*, Philadelphia, 1986, Saunders.

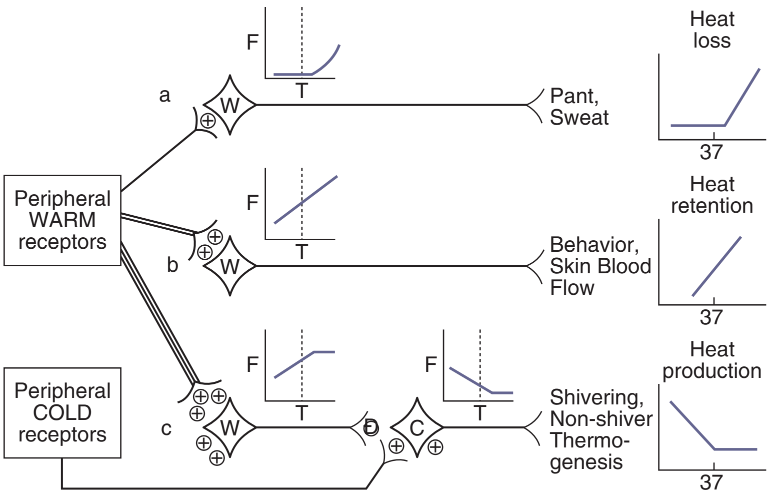

Figure 18-2 Central hypothalamic control of thermoregulation. Diagram shows neuronal control of various thermoregulatory responses. +, excitatory input; –, inhibitory input; dashed line, thermoneutral temperature; C, cold sensitive neuron; F, neuronal firing rate; T, hypothalamic temperature; W, warm sensitive neuron. (Redrawn fro Bouland JA, Dean JB: Temperature receptors in the central nervous system, *Ann Rev Physiol* 48:639-654, 1986.)

thermal load is also chemically mediated through the release of the neurotransmitters epinephrine and norepinephrine from the adrenal medulla to induce cutaneous vasoconstriction.[9]

A primary effect of cryotherapy is analgesia and the concomitant reduction of reflex muscle spasm. This may be caused by the decreased nerve conduction velocity that occurs when nerves are cooled. This relationship, known as the Q10 effect, is thought to be linear until 10° C, when neural transmission is blocked.[12,13] Nerve cooling also increases the duration of the refractory period, the time when a nerve cannot be stimulated by a second impulse.[14] Conversely raising the subcutaneous tissue temperature increases sensory nerve conduction velocity, with the greatest effect occurring during the initial 2° C temperature elevation.[15] Another mechanism postulated for the analgesic effect afforded by cryotherapy is that the cold receptors are overstimulated by cryotherapy, resulting in pain control at the spinal level by preventing pain transmission to higher centers via the spinal gate control theory of pain transmission.

Cryotherapy decreases muscle spasm through a number of possible mechanisms. Muscle spasm compromises venous return and promotes lactate accumulation and acidosis. The muscle spindle receptors and Golgi tendon organ receptors fire more slowly when cooled, although the effect is less pronounced on the Golgi tendon organs.[16] Sudden cooling has an excitatory effect on the muscle spindle, resulting in increased alpha motoneuron activity and increased muscle guarding. As the temperature continues to decrease, primary spindle afferent activity diminishes.[12] Muscle spasm, resulting from the pain-spasm-pain cycle, stimulates the static stretch response of the type II tissue afferents. Applying superficial heat until the temperature is above 42° C decreases the muscle spindle discharge rate while it increases the firing rate of the Golgi tendon organ.[16] This sequence of events results in a decrease in the firing rate of the alpha motor neuron. Thus cold may raise the threshold stimulus for muscle spindle activity, decreasing muscle spasm.[17]

Thermal effects on strength and endurance have also been observed by a number of authors. The force-velocity curve of muscle contraction shifts downward with decreasing temperature.[18] For a given concentric contraction velocity, cooler muscles have a lower force output. Interestingly, some studies suggest that proprioception, joint position, and balance remain largely unaffected by cryotherapy, whereas others indicate that joint position awareness is affected by joint cooling.[19-21]

Although most authors agree that changes occur in muscle torque production with cooling, it is less clear whether those effects are specific to a particular contraction mode (concentric acceleration or eccentric deceleration). One group of researchers reported that eccentric muscle action is augmented following cryotherapy,[22] whereas another group found increases in concentric, but not eccentric, torque.[23] Still others found no effects on peak torque, but increased muscle endurance following cooling.[24] Functionally, vertical jump performance, an indicator of lower extremity explosive power, is adversely affected by cryotherapy.[25,26] These findings suggest that caution must be used in making recommendations for performance following cryotherapy. Clearly the literature on this subject is undecided as to the specific effects of cold (both immediate and delayed) on muscle strength. These findings have the most implication for dogs who are engaged in work or competitive athletic pursuits. It also requires that the clinician understand the physiology and biomechanics of muscle actions and sport demands. Cold and exercise are an effective combination in the treatment of acute musculoskeletal dysfunction unless restrictions in flexibility are present.[27,28] Combining movement with cold allows for relatively greater pain-free exercise and assists in muscle pump activity to reduce acute injury-related edema.

Because of these effects, cold is most effective when applied during the acute phase of trauma, typically in the first 72 hours after injury or surgery. Cryotherapy may also be useful in chronic pain situations such as arthritic joints. With this condition it can be used to reduce pain, edema, collagenolysis, synovial inflammation, and joint destruction.[1,4,5] As a result of its analgesic effect, cryotherapy may permit decreased use of drugs for pain. This was noted in human medicine in a study examining numerous patients after anterior cruciate ligament surgery. Cryotherapy is currently used in postoperative treatment for swelling and pain, in musculoskeletal injury, muscle spasm, and after exercise to prevent edema or pain.

This thermal modality may also have a significant influence on the effectiveness of laser therapy, given its effects on blood flow and the saturation of blood chromophores in the tissue. Where tissue perfusion is reduced with cryotherapy, the absorption of laser energy may be improved, so that deeper structures may receive greater laser energy. This use of cryotherapy before laser therapy treatment could be useful for large dogs in which the target tissue is deeply located.[29]

Cryotherapy Applications

Cold application may be applied in several ways: cold water-circulating blankets, reusable ice packs, ice cubes wrapped in a towel, ice cups, cold immersion, cold compression devices, vapocoolant sprays, or contrast baths (Figure 18-3). Selection of the method depends on the desired effects, depth of penetration desired, stage of tissue healing, treatment area, and physiologic goals.

Cryotherapy may be achieved by the following physical mechanisms:

- Conduction (ice/cold packs, iced towels, ice massage, contrast bath, cold compression units)

Figure 18-3 Cryotherapy options (*clockwise starting top right*): Popsicle-type molds for ice massage, alcohol-water slush plastic bags, cotton wraps, commercial cold wraps, fluoromethane cold spray.

- Convection (cold baths)
- Evaporation (vapocoolant sprays)

Local application of cold may increase connective tissue stiffness (sometimes with decreased tensile strength) and may temporarily increase muscle viscosity (with decreased ability to perform rapid movement).

Generalized application of cold over the whole body, or large portions of it, causes decreased respiratory rate; generalized vasoconstriction in response to cooling of the hypothalamus; increased muscle spindle bias, which can increase spasticity (opposite to the effect of localized cold application); and increased muscle tone, which may be accompanied by shivering. Knowing this, if a patient has lower motor neuron flaccidity, cryotherapy may be used to temporarily facilitate muscle contraction.

For sanitary reasons and to improve patient comfort and prevent damage by cryotherapy to the treated tissues, a thin towel or other material (e.g., towel or pillowcase) is commonly placed between the cold source and the skin. It is very important to observe the skin for response to cold during the treatment. A normal response at the end of a cryotherapy treatment is reddening of the skin, and an abnormal response indicating that damage may have occurred is whitening or blanching of the skin. Because of skin pigmentation in some dogs, these color changes may not be apparent, so it may be better to err on the side of caution and reduce the length of application time if there is any question. The expected changes in sensation during a treatment follow the acronym IBAN: intense burning, aching, and numbness.[8]

Cryotherapy is a superficial treatment, but effects of up to 1 to 4 cm of tissue depth may be noted. The depth depends on the amount of adipose tissue and local blood flow. For these reasons, cryotherapy may be more effective in the distal area of the limbs where joint capsules, tendons, and ligaments are more superficial and there is less adipose tissue.[3,15,30]

Ice Packs/Cold Packs

Examples of ice packs/cold packs are crushed ice placed in a moist towel; a plastic bag wrapped in a moist towel; or water and rubbing alcohol of a 3 : 1 ratio in a sealed plastic bag and placed in a freezer (this should result in a slush-type consistency). Numerous types of cold packs are available commercially (Figure 18-4). The treatment time is generally 10 to 20 minutes for an application of cryotherapy.

The latent heat of fusion is lower for the alcohol-water mix than for water alone, therefore it performs similarly to crushed ice alone but at a lower temperature. Use caution with the water-alcohol mix because the potential for cold-induced injuries is greater. Never apply this type of cold pack directly to the skin. Instead, apply using a layer of toweling, either wet or dry (wet conducts cold more rapidly), between the cold pack and skin. Commercial cold packs may also be applied. Packs that freeze solid (e.g., DuraKold) perform better than frozen gel packs, which contain gelatin or antifreeze to make them more pliable (Figure 18-5).[4] Cold packs containing ethylene glycol antifreeze should not be used in animals because the contents are toxic if ingested. As with the alcohol-water mixture, use caution because the risk for frostbite is greater than when using a crushed ice bag (see Figure 18-5).

Bocobo et al studied the effects of cryotherapy on intraarticular temperature in dogs' knees.[5] Four treatment protocols were examined on 24 dogs: ice pack applied for (1) 5 minutes, (2) 15 minutes, (3) 30 minutes, and (4) local ice bath immersion for 15 minutes. Intraarticular temperature decreased in a time-dependent fashion, with temperatures decreasing 2.2° C, 4.1° C, and 6.5° C after 5, 15, and 30 minutes of ice pack application, respectively. Temperatures continued to drop for a short period of time after the ice pack was removed, and required 21.7, 33.2, and 27.8 minutes to return to pretreatment temperatures, respectively. Ice bath immersion resulted in the greatest decrease in joint temperature, 20.2° C and had the greatest rewarming time of 59.8 minutes. Interestingly, rectal temperature did not decrease with 5 or 15 minutes of ice pack application, and only decreased 0.5° C with 30 minutes of treatment. Ice bath immersion caused a 1.6° C drop in rectal temperature, however. These data indicate that under normal circumstances, local application of ice packs results in tissue cooling in a time-dependent manner with minimal effects on core body temperature. Furthermore, the cooling effect appears to plateau after 30 minutes, suggesting that treatments longer than 30 minutes offer no further benefit in regards to intraarticular temperature decrease. Immersion in an ice bath results in more rapid and greater cooling than ice pack treatment. There was a prolonged rewarming period in all tissues after the removal of cryotherapy, and these results should be considered before beginning cryotherapy treatment.

Vannetta et al. also examined the effects of a 20-minute cold pack treatment to dogs on skin and muscle temperature as well as intramuscular blood flow.[31] Temperatures and blood flow ratings were recorded for each minute of the application and for 80 additional minutes following the removal of the cold pack (Figure 18-6). They found that the temperature of the skin and subcutaneous tissues decreased to a greater degree than did the tissues at depths of 1 and 3 cm. The temperature of the superficial tissues rose sharply following removal of the cold pack but the temperature of the deeper tissues continued to decrease for 10 minutes and remained decreased for 60 minutes before beginning to return to baseline. Blood flow was decreased in the superficial tissues and remained decreased after the cold pack was removed. Blood flow at 1- and 3-cm depths, however, were initially decreased but began to rise before the treatment ended and remained above baseline levels for the 80 minutes following the cold pack removal. This study

Figure 18-4 **A,** Example of a cryotherapy treatment using a cold pack wrapped around a dog's hind limb. **B,** Canine stifle icer. **C,** Carpal icer.

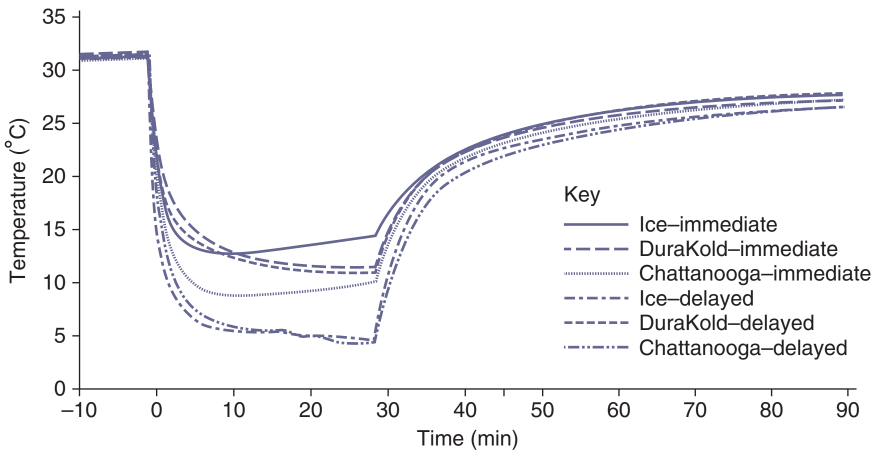

Figure 18-5 The effect of the heat of fusion can be seen in this comparison of several forms of cold packs. The crushed ice and DuraKold packs cool the skin more than the Chattanooga frozen gel pack. The effect is even greater when application is delayed for 20 minutes after the packs are removed from their cooling devices. (Knight KL, A re-examination of Lewis' cold-induced vasodilation in the finger and ankle. *J Athl Train* 15:238-250, 1980.)

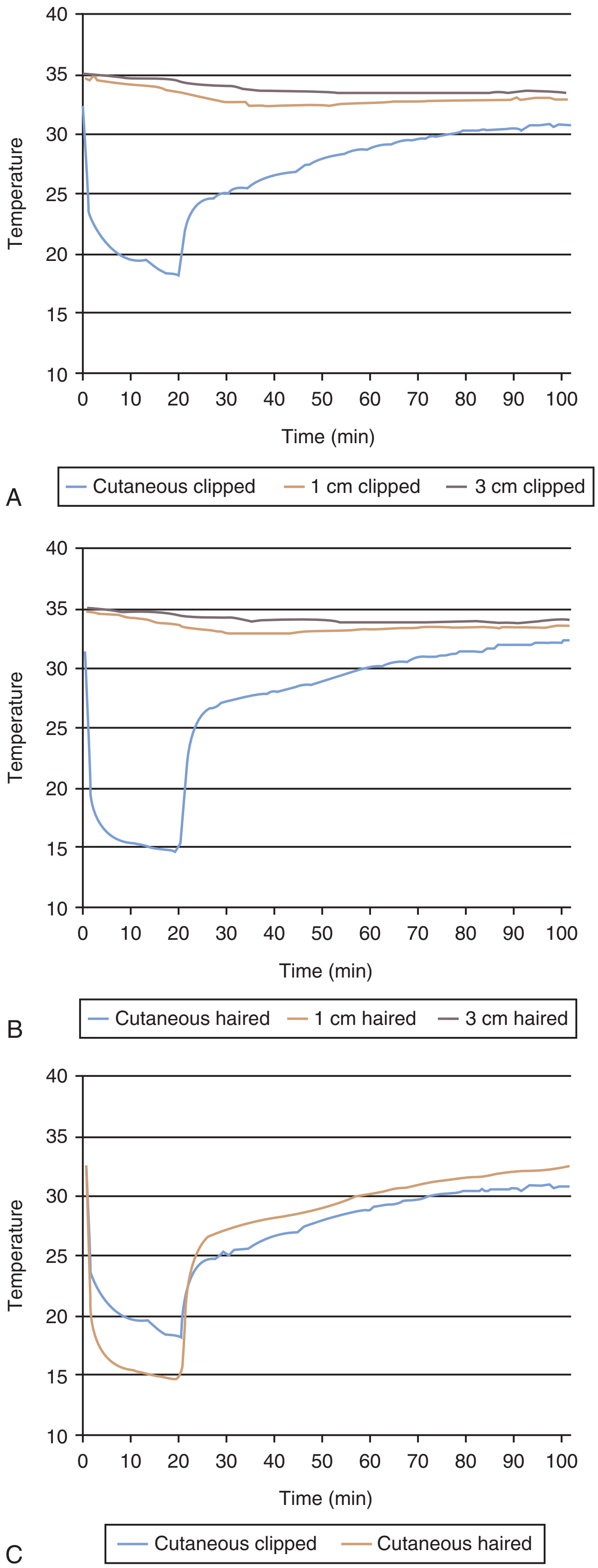

Figure 18-6 **A,** Temperature of caudal thigh muscles during cryotherapy application for 20 minutes and rewarming in dogs with the hair clipped. Note that the most dramatic temperature changes occur in the skin. Changes in the deeper tissues occur slower, but the effects are more prolonged, presumably as a result of heat transfer from regions of higher temperature to lower temperature. **B,** Temperature of caudal thigh muscles during cryotherapy application for 20 minutes and rewarming in dogs with hair. Note that the most dramatic temperature changes occur in the skin. Temperature decrease is greater in dogs with hair during cryotherapy application, and rewarming occurs more quickly, likely because of the insulating effect of the hair. Changes in the deeper tissues occur slower, but the effects are more prolonged, presumably as a result of heat transfer from regions of higher temperature to lower temperature. **C,** Temperature of the skin over the caudal thigh muscles during cryotherapy application for 20 minutes and rewarming in dogs with the hair clipped or with hair intact. Note that greater changes occur in haired dogs, suggesting that hair provides an insulating effect.[31]

concluded that the temperature of deeper tissues may remain decreased for up to 70 minutes following a cold pack treatment. Also blood flow increases following the removal of a cold pack in deeper tissues, perhaps in an attempt to rewarm the tissues back to their baseline levels.[31]

The effects of cold −16° C (3° F) compresses were studied on epaxial muscles in dogs. The gel compress was applied for 5, 10, and 20 minutes in random order and temperature changes at 0.5, 1.0, and 1.5 cm of tissue depths were measured. At 10 minutes, the cold decreased tissue temperature at 0.5, 1.0 and 1.5 cm depths 7° C, 4.7° C, and 4° C, respectively. At 20 minutes the cold further decreased tissue temperature to 8.2° C, 6.5° C, and 4.7° C at the respective depths. The dogs were of ideal body condition making the results a good guide, but not applicable to all dogs. The differences between 10- and 20-minute applications were significant but applications of 10 minutes were sufficient for cooling.[32]

Cold-Compression Units

Cold-compression units alternately pump cold water and air through separate tubing in a sleeve wrapped around a patient's limb. The temperature of the water is usually set between 2° and 10° C (35 to 50° F) to provide cooling. Compression is applied by intermittent inflation of the sleeve with air. Units manufactured for humans may be noisy, and the sleeves for human use do not typically conform well to dogs. Commercially available units are now made for use in dogs (Figure 18-7) and can be a highly effective treatment during the acute phase of inflammation or tissue healing.

Cold Immersion

Cold immersion treatments consist of immersion of an affected body part in a container of cold water. The temperature of the water for this type of treatment is usually 2° to 16° C (35° to 60° F), and the treatment generally lasts for 10 to 20 minutes. Cold water immersion results in the greatest decrease in tissue temperature because the largest surface area is in contact with the cold source. This is a difficult form of cryotherapy to use in veterinary medicine because of compliance issues, and it requires the patient to stand in the immersion vessel. The target tissues should be the only areas immersed in the water. Cold water immersion should not be used in the presence of large wounds because of the risk of infection, and delayed healing may result.[26]

Contrast Baths

Contrast baths are applied by alternate immersion of an area in warm and cold water. This modality produces alternating vasodilatation and vasoconstriction and is sometimes used to decrease edema. Contrast baths are appropriate not only in the acute phase, but during the early subacute phase of tissue healing as well. During the acute phase of tissue healing, the cycles of alternating immersion should have greater emphasis on cold water. An example of this would be immersion in the cold water for 3 minutes and then immersion in warm water for 1 minute. As the stage of tissue healing progresses, the times spent in warm water may increase as the time spent in the cold decreases. Treatment time for this method of cryotherapy may last from 15 to 30 minutes. Contrast bath treatment for edema should end with cold. The varying sensory stimulus is also thought to promote pain relief and desensitization. This treatment should be considered for conditions such as chronic edema, subacute trauma, sprains, strains, and tendonitis. Alternating hot and cold packs has also been proposed as an alternative to contrast baths.

Ice Massage

Freezing water in cups or any other form of cylinder can be used to form an ice massage medium (Figure 18-8). To

Figure 18-7 Game Ready cold compression device with stifle sleeve, http://www.gamereadycanine.com.

Figure 18-8 Ice massage being performed on dog's hind limb.

apply the massage the therapist holds onto the cup, exposes the ice surface, and puts it in direct contact with the patient's skin. Treatment time is generally 5 to 10 minutes or until the affected area is erythematous, slightly pink or red, and numb. This technique can be useful for small, irregular areas. For large areas, a cold pack is more effective regarding efficient use of time.

Vapocoolant Sprays

These sprays are liquids used to achieve a brief and rapid cooling effect when applied to the skin. The spray absorbs heat as it hits the skin, and the increase in temperature causes it to change from a liquid into a vapor. To use the vapocoolant spray to treat trigger points, place the affected muscle in a position of slight stretch and spray from proximal to distal at a rate of 10 cm/sec from a distance of 30 cm (12 inches). Repeat four or five times until the affected area has been covered. It is important to cover the entire length of the muscle with the spray, and the muscle should be stretched during and after the application. This type of treatment is frequently applied directly over the skin just superficial to a trigger point in human populations.[33] This can be a difficult modality to apply to dogs because of the hair covering the skin surface. Also, some of the commercially available sprays may be harmful to animals if they contact an animal's eyes or are ingested.

Treatment, Duration, and Frequency

Cryotherapy treatment times depend on the modalities chosen and the treatment area, but the application time is usually 15 to 20 minutes and may be repeated throughout the day. A common method is to repeat the cold application every 2 to 4 hours during the first 24 to 48 hours after the injury, or treat 10 to 15 minutes on and 10 to 15 minutes off twice. An ice treatment should not be applied for more than 20 minutes per session to avoid tissue damage. Local hypothermia should never exceed 30 minutes per treatment session for the same reason, as well as to avoid stimulating unwanted local vasodilatation and edema formation.

Precautions and Contraindications

The therapeutic effect of cryotherapy is believed to occur when the tissue temperature reaches 15 C° to 19° C, but this is dependent on the tissues normal temperature. Previous regions of frostbite should be avoided. Although it may be difficult to check skin color of dogs for redness or blanching because of pigmentation and hair coat, it should be done after the first 5 to 10 minutes to be certain that continued treatment is safe. Apply cryotherapy with caution over superficial nerves; open wounds (because of the vasoconstriction that occurs); fractures; areas where there may be sensory or motor nerve impairment; previous frostbite; in patients with hypertension (cold may increase blood pressure with systemic application); to areas with decreased or absent sensation; and in very old or very young patients.[34]

Cryotherapy Applied with Coaptation Devices in Place

Dogs and cats frequently have external coaptation devices, such as bandages or casts, placed over limbs following injury or surgery. Because cryotherapy is useful during the acute postinjury phase, it would be beneficial to apply cryotherapy over coaptation devices to achieve the beneficial effects of the cryotherapy without the need to remove and reapply coaptation devices. In the case of cast application following fracture treatment, it would not be practical or advisable to remove and replace a cast several times daily.

In general it is believed that a cast or bandage prevents or reduces the effects of cold on the skin when applied over a cast or bandage. Several studies have addressed the use of cryotherapy over casts or bandages in people, and although specific studies are not available in animals, these findings may serve as guidelines for use in animals.

One study evaluated the magnitude of temperature changes at the skin of the ankle after the application of frozen ice packs to the surface of various casts and bandages both in normal and swollen ankles.[35] A Robert Jones bandage, elastic support bandage, a below-knee plaster cast, or synthetic below-knee cast were applied to patients, and two frozen ice packs were placed around the cast or bandage. Although skin temperature under dressings and casts decreased significantly relative to the baseline temperatures with application of two frozen ice packs in all groups, less skin temperature decrease occurred with the Robert Jones bandage (32° to 23° C) as compared with casts (31° to 16° C with fiberglass casts). In addition, the length of time to achieve the lowest skin temperature required 46 minutes compared with 36 minutes for the Robert-Jones bandage and fiberglass cast, respectively. Furthermore this study only evaluated skin temperature decrease, so the effects on deeper tissues are unknown.

A related study found similar results.[36] Over a 90-minute period of ice pack application, the average final skin temperature was 16.5° C under a plaster cast, 18.8° C under a synthetic cast, 21.2° C under an Ace wrap dressing, and 28.7° C under a Robert-Jones dressing. The authors concluded that ice application effectively decreased skin temperatures through the plaster cast, synthetic cast, and Ace wrap dressing, but the Robert-Jones dressing prevented adequate cooling. Other studies have similarly concluded that a synthetic or plaster cast does not eliminate tissue cooling when crushed ice packs are applied to the surface of the casts, but the time required for maximum cooling is approximately 1 hour,[37] and bulky dressings inhibit adequate cooling.[38]

Postoperative Cryotherapy

A recent study compared cold compresses, a modified Robert Jones bandage, cold compresses plus a Robert

Jones bandage, and microcurrent electrical stimulation in combination with a Robert Jones bandage. The use of cold compresses alone or with a modified Robert Jones bandage, or using microcurrent electrical stimulation in combination with a modified Robert Jones bandage decreased soft tissue swelling over 72 hours more than bandaging alone after extracapsular repair of CCLR.[39] A similar study in patients following TPLO surgery found that cold compression therapy decreased signs of pain, swelling, and lameness, and increased range of motion during the first 24 hours after surgery.[40]

Heat Therapy

Few therapeutic procedures have been historically used for as long as heat; it is one of the oldest forms of physical therapy. Superficial heating agents penetrate to tissue depths up to approximately 2 cm, whereas deep heating agents elevate tissue temperature to depths of 5 cm. Although skin temperature may increase 10° C or more, tissues at 1 cm in depth are typically raised less than 3° C and tissues at 2 cm less than 1° C.

Physiologic Effects of Heat

The effects of heat are opposite those of cold, except that heat and cold both may relieve pain and muscle spasm. With an increase in tissue temperature, there is a sedative and analgesic effect, increased leukocyte migration into the heated area, increased metabolic rate, and arteriolar dilation that results in increased capillary blood flow (which can promote edema) and decreased blood pressure (if heat is applied for a prolonged time or over a large surface area).[1,34,41] Other benefits of superficial heat may include reducing edema (however, heat should not be used during acute inflammatory periods because edema may be exacerbated) and increasing patient comfort. The most important beneficial physiologic effects of heat are:

- Increased circulation (vasodilation)
- Pain relief
- Increased soft tissue extensibility
- Relaxation of muscle spasm

Heat causes vasodilation and increases the rate of blood and lymph flow, thus potentiating the reabsorption of edema, which helps to remove tissue metabolites, and increase tissue oxygenation and metabolic rate. Normally the metabolic rate increases two to three times for every 10° C (18° F) increase in temperature. Increased enzymatic and metabolic activity has been observed in tissues at 39° to 43° C (102° to 109° F), and the activity rates continue to increase until 45° C (113° F). Beyond this temperature, the protein constituents of enzymes begin to denature and enzyme activity rates decrease, ceasing completely at about 50° C (122° F).[17,34] Any increase of enzymatic activity results in an increased rate of cellular biochemical reactions. This can increase oxygen uptake and accelerate healing but may also increase the rate of destructive processes.

Thermotherapy may cause vasodilation by a variety of mechanisms, including direct reflex activation of the smooth muscles of the blood vessels by the cutaneous thermoreceptors, indirect activation of local spinal cord reflexes by the cutaneous thermoreceptors, or by increasing the local release of chemical mediators of inflammation (histamine, prostaglandin, bradykinin). These cause an increase in capillary permeability as well.

Superficial heating agents stimulate increased activity of the cutaneous thermoreceptors. It is proposed that transmission from cutaneous thermoreceptors, via their axons, communicate directly to nearby cutaneous blood vessels to cause the release of bradykinin and nitrous oxide. Bradykinin and nitrous oxide act as vasoactive mediators stimulating relaxation of the smooth muscles of the blood vessel walls causing vasodilation. This vasodilation occurs locally, in the area where the heat is applied. Cutaneous thermoreceptors also project, via dorsal root ganglia, to synapse with interneurons in the dorsal horn of the gray matter of the spinal cord. These interneurons synapse with sympathetic neurons in the lateral gray horn of the thoracolumbar segments of the spinal cord to inhibit their firing and thus decrease sympathetic output. This decrease in sympathetic activity causes a reduction in smooth muscle contraction, resulting in vasodilation both at the site of heat application and in the cutaneous vessels of the distal extremities. This distal vasodilative effect of thermotherapy may be used to increase cutaneous blood flow to an area where it is difficult or unsafe to apply a heating agent directly. The vasodilation of the vessels of the distal extremities can promote the reabsorption of edema, but in the same way can promote edema if applied during times of acute inflammation.[34]

Local application of heat increases sensory and motor nerve conduction velocity. An increase or decrease in conduction velocity reduces pain and muscle spasm. Nerve conduction velocity increases by approximately 2 m/sec for every 1° C (1.8° F) increase in temperature.[34] The clinical implications of these effects are not well understood. They may contribute to reduced pain perception or improved circulation that occurs in response to the increasing tissue temperature. Nerve firing rate (frequency) also changes in response to changes in temperature. Elevation of muscle temperature decreases the firing rate of type II muscle spindle efferents and gamma efferents and increases the firing rate of type Ib fibers from Golgi tendon organs. These changes in nerve firing rates are thought to contribute to a reduced firing rate of alpha motor neurons and thus, to a reduction in muscle spasm. The decrease in gamma neuron activity causes the stretch on the muscle spindles to decrease, reducing afferent firing from the spindles. The decreased spindle afferent activity results in decreased alpha motor neuron activity and thus, relaxation of muscle contraction.[11,22]

Another effect of heat is to relieve pain. Several studies demonstrate that the application of local heat can increase an individual's pain threshold. The inhibitory control of pain at the spinal cord level was explained in the gate control theory of Melzack and Wall in 1965.[42] This theory suggests that the perception of pain is not a direct result of activation of nociceptors, but instead is modulated by the interaction between different neurons, both pain-transmitting and non–pain-transmitting. The theory asserts that activation of nerves that do not transmit pain signals can interfere with signals from pain fibers and inhibit an individual's perception of pain. The gate control theory suggests that sensory inputs of pain are modulated through ascending and descending pathways in the central nervous system. Descending neural pathways potentiate or attenuate pain signals influencing the amount of neurotransmitter released by the incoming neurons or by changing the sensitivity of the ascending nerves in the spinal cord to these neurotransmitters. Afferent pain-receptive nerves, those that bring signals to the brain, are comprised of at least two kinds of fibers: a fast, relatively thick, myelinated A delta fiber that carries messages quickly with intense pain and a small, unmyelinated, slow C fiber that carries the longer-term throbbing and chronic pain. Large-diameter A beta fibers are non-nociceptive (do not transmit pain stimuli) and inhibit the effects of firing by A delta and C fibers.

The peripheral nervous system has centers at which pain stimuli can be regulated. Some areas in the dorsal horn of the spinal cord that are involved in receiving pain stimuli from A delta and C fibers also receive input from A beta fibers. The non-nociceptive fibers indirectly inhibit the effects of the pain fibers, "closing a gate" to the transmission of their stimuli. In other parts of the laminae, pain fibers also inhibit the effects of non-nociceptive fibers, "opening the gate." An inhibitory connection may exist with A beta and C fibers, which may form a synapse on the same projection neuron. The same neurons may also form synapses with an inhibitory interneuron that also synapses on the projection neuron, reducing the chance that the latter will fire and transmit pain stimuli to the brain. The inhibitory interneuron fires spontaneously. The C fiber's synapse may inhibit the inhibitory interneuron, indirectly increasing the projection neuron's chance of firing. The A beta fiber, on the other hand, forms an excitatory connection with the inhibitory interneuron, thus decreasing the projection neuron's chance of firing. (Like the C fiber, the A beta fiber also has an excitatory connection on the projection neuron itself.) Thus, depending on the relative rates of firing of C and A beta fibers, the firing of the non-nociceptive fiber may inhibit the firing of the projection neuron and the transmission of pain stimuli. The gate control theory thus explains how a stimulus that activates only non-nociceptive nerves can inhibit pain. One area of the brain involved in reduction of pain sensation is the periaqueductal gray matter that surrounds the third ventricle and the cerebral aqueduct of the ventricular system. Stimulation of this area produces analgesia by activating descending pathways that directly and indirectly inhibit nociceptors in the laminae of the spinal cord. It also activates opioid receptor–containing parts of the spinal cord.[27,34,42] Therefore the control of pain produced by thermotherapy may be caused by gating of pain transmission by activation of cutaneous thermoreceptors, but may indirectly be the result of improved healing, decreased muscle spasm, or reduced ischemia.

Muscle strength and endurance may decrease during the initial 30 minutes after the application of deep or superficial heating agents. It is proposed that this initial decrease in muscle strength is the result of changes in the firing rates of type II muscle spindle efferent, gamma efferent and type Ib fibers from Golgi tendons organs caused by heating of the motor neurons. Beyond 30 minutes after the application of heat and for the next 2 hours, muscle strength gradually recovers and then increases to greater than pre-treatment levels.[11]

Increasing the temperature of soft tissue increases its extensibility. When soft tissue is heated before stretching, the length gained is maintained for a greater amount of time after the stretching force is applied, less force is required to achieve the increase in length, and the risk of tissue damage is reduced. If heat is applied to collagenous soft tissue such as tendon, ligament, scar tissue, or joint capsule before prolonged stretching, plastic deformation, whereby the tissue increases in length and maintains most of the increase after cooling, can be achieved. In contrast, if collagenous tissue is stretched without prior heating, elastic deformation, whereby the tissues increase in length while the force is applied but lose most of the increase when the force is removed, may occur.[34,43] For this reason, thermotherapy can be used clinically to increase joint ROM and decrease joint stiffness. Both of these effects are, in fact, the result of the increase in soft tissue extensibility that occurs with increased soft tissue temperature. Increasing soft tissue extensibility contributes to increasing joint ROM, especially if stretching is performed immediately after the application of heat or during it. If the tissues are allowed to cool before being stretched, the effects of prior heating on tissue extensibility are lost.[34]

Decreased muscle spasm may be a result of relieving the ischemia caused by the chronic contraction of a muscle and partial occlusion of the blood vessels. Superficial heating also decreases muscle spindle activity by decreasing gamma efferent activity, which may also be responsible for the decrease in spasm as well as breaking the pain-spasm cycle.[34]

Heat Applications

Superficial heating agents typically heat the skin and subcutaneous tissues to a depth of 1 to 2 cm. Although skin temperature may increase 10°C or more, tissues at 1 cm in depth are typically raised less than 3°C and tissues at 2 cm less than 1°C. The tissues are usually heated for 15 to 20 minutes to promote hyperemia.[34]

Heat therapy may be transmitted by:

- Conduction (hot packs, paraffin, hydrotherapy)
- Convection (hydrotherapy, air temperature)
- Radiation (UV lamps, infrared)

Heat therapy (Figure 18-9) may be used to relieve pain, relax muscle spasm, increase the blood supply to an area, and increase soft tissue extensibility. Subacute and chronic inflammation or traumatic conditions such as contusions, sprains, strains, and myositis can also benefit from heat therapy.

Superficial heating agents may include: hot packs, heat wraps, hosing with warm water, whirlpools, paraffin baths, circulating warm water blankets, electric heating pads, and infrared lamps. The last two are considered to have a higher risk of burns in animals. Another common form of superficial thermotherapy is heated beds for dogs, which have been used for comfort or conditions such as arthritis.

Hot Packs

Different forms of hot packs include heated sacks with a canvas covering that may be filled with cracked corn, beans, bentonite (hydrophilic silicate gel), or other inert materials; electric heating pads; damp microwaved towels; or circulating warm water blankets. There are different kinds of hot pack shapes and sizes designed to fit different areas of the body, but it is important to consider the depth of tissues that may be affected by the modality in relation to the tissues intended to receive benefit.

Figure 18-9 Example of a hot pack placed over a dog's hip.

Hot packs may be reusable and usually can maintain an elevated temperature for 20 to 30 minutes. The therapist should be certain that the hot pack is not too warm. A good rule of thumb is that if the hot pack is comfortable when it is placed on the back of the therapist's neck, it should be safe for the patient. Placing the therapist's hands under the hot pack and against the animal's skin is also a good way to check for appropriate temperature. To further avoid burns to the treated tissues, the therapist should place a towel between the heat source and the skin. It is very important to observe the skin's response to the treatment and whenever the skin feels very warm or hot or looks red, another layer of toweling should be added between the pack and skin. If overheated, dogs with normal sensation will try to move away from the pack but dogs with impaired sensation may not realize the area is overheating and are especially susceptible to burns. Hot packs are a relatively safe form of thermotherapy because they cool during treatment, minimizing the risk of burn.

The effects of warm 47°C (116°F) compresses were studied on epaxial muscles in dogs. The gel compress was applied for 5, 10, and 20 minutes in random order and temperature changes at 0.5, 1.0, and 1.5 cm tissue depths were measured. The compress significantly increased tissue temperature of the lumbar region (>2°C) at 0.5 and 1.0 cm depths, but heating was minimal at 1.5 cm depth. The compresses in this study were not as hot as many commercial products and the dogs were of ideal body condition making the results a good guide, but not applicable to all dogs and all hot packs.[44]

Warm Water

Warm water is another way of applying superficial heating. Water may be applied directly to the area being treated, towels heated with warm water may be used, or the affected segment may be immersed in a warm water bath or a whirlpool. If the last method is chosen, be mindful that systemic heating with a warm bath may decrease blood pressure and increase the heart rate.[34] Whirlpools have the advantage of also providing increased hydrostatic pressure to submerged body parts, which may reduce edema. Increased hydrostatic pressure helps to increase lymphatic and venous flow from a distal to proximal direction. The temperature of a whirlpool is based on the needs of the individual animal. For example, patients with chronic conditions may be treated with warmer water than patients with more acute disorders. When using an underwater treadmill in conjunction with thermotherapy, the temperature may range from approximately 27° to 35°C (80° to 95°F), depending on how vigorously the dog is exercising. The animal may be immersed in water at approximately 30°C after exercise to have a moderate heating effect, which continues to increase blood circulation and promote the removal of lactic acid and other metabolites produced following intense exercise.[34]

Paraffin Baths

Warm, melted paraffin is another method of thermotherapy. For this application, the paraffin is mixed with mineral oil in a 6 : 1 or 7 : 1 ratio of paraffin to oil to reduce the melting temperature of the paraffin from 54° C (129° F) to between 45° and 50° C (113° to 122° F). Paraffin can safely be applied at this temperature because of its low specific heat and thermal conductivity. Paraffin is usually used for heating the distal extremities because it can maintain good contact with these irregularly contoured areas.[34] Treatment time for this method is 15 to 20 minutes. The use of paraffin baths in veterinary medicine is messy, and therefore uncommon.

Infrared Lamps

Infrared lamps emit electromagnetic radiation in the frequency range that gives off heat when absorbed by matter. The tissue temperature increase produced by infrared lamp radiation is directly proportional to the amount of radiation that penetrates the tissue. This is related to the power and wavelength of the radiation, the distance of the radiation source from the tissue, and the absorption coefficient of the tissue. Higher power infrared delivers more radiation to the skin.[15] Infrared lamps emit infrared light that penetrates the skin to promote increased blood flow and circulation. This method of thermotherapy is used to warm large areas of the body. The lamps are positioned 30 to 40 cm from the affected area. The therapist should place his or her hand under the heat lamp for several minutes at the desired height to be sure that the temperature is comfortable. Because of the dog needing to remain stationary during treatment, this method of heating is uncommon.

Treatment Duration and Frequency

The duration and frequency of thermal treatment depend on the severity of the injury, the stage of tissue healing, the area of the injured tissues, and the desired outcome. Heat treatments generally last from 15 to a maximum of 30 minutes and may be repeated three or four times daily. If white areas (due to rebound vasoconstriction) or red mottled areas appear on the skin, the treatment should be stopped.[30] The most desired effects of heat are achieved when the temperature is increased between 2° and 4° C in the tissues. Tissue temperatures greater than 45° C can be painful and cause irreversible tissue damage.[18] Hot packs have their greatest application in combination with other physical procedures. They may be applied before or even during massage, exercise, or electrical stimulation. It is recommended that heat be applied no earlier than 48 hours after an injury, because earlier application may potentiate swelling. The treatment times depend on which method is chosen.[34]

Precautions and Contraindications

Electric heating pads and infrared lamps are forms of thermotherapy that have a higher risk of burns. Electric heating pads should never be placed under an anesthetized or heavily sedated animal or one with decreased superficial skin sensation. In general animals should never be left unattended during treatment and the skin should be monitored frequently.

The patient should be protected from burns by insulating hot packs with towels and by examining the skin every few minutes. If the skin feels excessively hot, more towels should be added to the insulation layer or the hot pack should be removed and allowed to cool before reapplying. Heat therapy should be modified or abandoned if it causes pain or there is an exacerbation of pathologic signs.

When hot packs are applied to an infected site or abscess, the suppurative process is enhanced. Because of the increased circulation, more phagocytes and antibodies enter the site, which is a positive response. Too much heat may increase pain in these cases, however, and it is better to use packs that are only moderately warm.

Heat promotes tissue extensibility of joint capsules, tendons, and scar tissue and decreases joint stiffness, so precautions must be taken to not overstress these tissues because joints are less stiff and more mobile during treatment. Although thermotherapy may reduce pain of any etiology, it is not recommended as a treatment for pain caused by acute inflammation because an increase in tissue temperature may aggravate other signs of inflammation, including heat, redness, and edema.[34]

Precautions are needed for patients that are pregnant, extremely obese, have impaired circulation, poor thermal regulation, or cardiac insufficiency. Also, animals with bursitis, tendonitis, impaired circulation, or noninflammatory edema have decreased resistance to heat and should be given heat therapy with caution.[45] The natural thermoregulatory mechanisms are less efficient in very young animals. In older patients, the temperature-regulating systems may function poorly or the patient's sensitivity to pain may be impaired. The lessened cardiovascular and respiratory reserves in older animals may cause them to be less tolerant of heat.[45]

Local application of heat is contraindicated in the following situations: active bleeding, acute inflammation, thrombophlebitis, cardiac insufficiency, fever, malignancy, presence of swelling or edema, and poor body heat regulation.[34] It should be used with extra caution in cases of decreased or impaired sensation and in cases of decreased or impaired circulation in the area to be treated (to avoid overheating). If there is a wound and heat cannot be applied to this area, a hot pack may be applied to a remote area with an excellent blood supply in the hope that reflex vasodilatation will occur at the treatment site, which may promote better wound healing.

Conclusion

Heat therapy and cryotherapy are powerful adjuncts in the rehabilitation of musculoskeletal injuries in dogs. Successful management depends on an accurate assessment of the dog's presenting problems at the beginning of each treatment session. Some effects and clinical indications for the use of cryotherapy and heat therapy are the same, but others are very different. There are some situations in which either may be used, and others in which only one would be appropriate. Careful consideration should be taken to choose the proper therapeutic modality to achieve an appropriate effect and understand how the modality chosen may facilitate or hinder a positive outcome.[45,46]

Cryotherapy and cryokinetics are most useful during the acute inflammatory stages of tissue healing. Two recent papers on cruciate repairs have demonstrated the effectiveness of cryotherapy to minimize inflammation.[39,40] Because tissue healing is initially established with relatively weak tissue strength, it is susceptible to injury from exercise that is too aggressive or from changing from cold to heat too early, thus exacerbating the inflammatory response. After new tissue has been established and the tensile strength of the tissue continues to increase with healing, the emphasis shifts to promoting a mobile scar, normal range of motion, and normal flexibility. In patients with limitations in these areas, concurrent application of superficial heat and flexibility exercises results in beneficial plastic changes to the tissues that may be relatively permanent. Superficial cold and heat modalities may be a very useful adjunct in a rehabilitation program to facilitate nature's ultimate therapeutic modality, controlled exercise.

Selection of the appropriate modality depends largely on an understanding and accurate assessment of the stage (acute inflammatory, proliferative, or tissue remodeling) of tissue healing, an accurate clinical assessment of the dog's functional abilities, the established treatment goals, and continued reevaluation as the patient's (and tissues') status changes.

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19 Therapeutic Ultrasound

David Levine and Tim Watson

The aim of physical rehabilitation is to reestablish normal function using modalities such as heat and electrical stimulation, mobilization, and therapeutic exercise.[1] Superficial heating agents penetrate soft tissues to a depth of approximately 1 cm. Deep heating agents can elevate tissue temperatures at depths of 2 cm or more. Deep heating agents commonly used include therapeutic ultrasound (US) and diathermy. Although diathermy is an effective deep heating modality, it requires the patient to remain stationary for 15 to 20 minutes and is not practical in dogs.

Therapeutic US is considered an effective treatment modality for rehabilitating musculoskeletal conditions such as restricted range of motion (ROM) resulting from joint contracture, pain and muscle spasm, and wound healing.[2] There are advantages and disadvantages of US therapy (Box 19-1). Many protocols for the administration of US are based on tradition or extrapolated from basic science research and remain to be tested in controlled clinical trials.[3-6] This chapter is based on a review of current literature pertaining to the basic research findings (in human and dogs) and the clinical application of US therapy in dogs.

Physical Principles

Energy within a sound beam decreases as it travels through tissue, because of scatter and absorption. Scattering is the deflection of sound out of the beam when it strikes a reflecting surface. Absorption is the transfer of energy from the sound beam to the tissues. Absorption is high in tissues with high protein content[7,8] and relatively low in adipose tissue.[7,9]

For more detailed information regarding the components of US machines, piezoelectric effects, sound generation, and sound propagation through tissues, the reader is referred to other sources.[10,11]

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Coupling Techniques

The therapist must first consider the site of soft tissue involvement, such as muscle, joint capsule, tendon, bursa, or scar tissue, and position the animal appropriately. Positioning is a compromise between patient comfort and access to the target tissue.

The US beam is attenuated in air and reflected at air-tissue interfaces. Consequently a coupling medium must be placed between the sound head and the skin.[12] Methods for direct coupling, immersion, and use of coupling cushions have been described for dogs.[13]

Direct coupling is preferred when the surface to be treated is relatively flat and is larger than the transducer head surface. A layer of water-soluble US gel is spread over the skin, and the sound head is placed in contact with the gel. Elimination of as much air as possible maximizes the US energy entering the tissues. If there are bony prominences in the treatment area or the surface is very small, smaller applicators (transducer heads) are preferable.

Commercially prepared water-soluble US gels are the best and most practical coupling method. The gel may be preheated. Coupling agents that are not recommended include[1] substances that may irritate or penetrate through the skin[2]; electroconductive gels, such as for electrocardiography or electromyography, which contain salts that can damage the transducer face[3]; lanolin-based compounds, which would cause heating of the transducer head with poor transmission to tissue[8]; and mineral oil,[4] which is messy to remove.[13] Over-the-counter creams and lotions should not be used because they generally do not provide effective transmission of US.

Other methods of coupling have been used. However, research results on these indirect coupling methods are inconclusive, and further investigation is warranted.[14] The underwater method was popular before smaller transducer heads became available.[8] Immersion should be considered only when the surface to be treated is so uneven that direct contact is too difficult. The part of the body to be treated, usually a distal limb, can be immersed in a container of tap

Box 19-1 Advantages and Disadvantages of Therapeutic Ultrasound

Advantages

Produces local heating of deeper tissues
Treatment time is short, approximately 10 minutes

Disadvantages

The dosage is difficult to monitor
Transducer contact with the skin may aggravate preexisting tenderness of the treated area

water at room temperature. The water and the animal's skin must be clean.[15] Water in a whirlpool is not recommended if it has been agitated because air bubbles form in the water. Deionized, degassed water is not necessary, but the water may be boiled and cooled or allowed to stand to remove air bubbles. Even so, small air bubbles tend to accumulate on the transducer face and the skin during treatment. The therapist can wipe these off quickly, but should try to minimize his or her own exposure. With regard to container type, metal containers reflect some of the US beam, which could increase the intensity in areas near the metal; rubber or plastic do not cause as much reflection. During treatment, the transducer should be held underwater 0.5 to 3.0 cm from the skin surface. The intensity may be increased by 0.5 W/cm^2 to compensate for absorption by water.[7]

Immersion has not compared favorably with direct coupling in several studies that examined the tissue temperatures achieved with these two methods. One study compared immersion to topical US gel on temperature rise at a depth of 3 cm in the gastrocnemius muscle of humans.[12] The treatment consisted of continuous US at 1.5 W/cm^2 for 10 minutes, with the sound head moved at a speed of 4 cm per second. With direct coupling, tissue temperature increased 4.8° C compared with 2.1° C with immersion. In another study the lateral epicondyle of pigs was treated with the transducer in direct contact with the skin and with the limb immersed in a water bath with the applicator held 2 cm from the skin surface. The temperature in the tendons of the extensor muscles originating at the lateral epicondyle rose into the therapeutic range (an increase of 2° or more) with direct coupling, but not with immersion.[16]

Another alternative to direct coupling is the use of a coupling cushion.[13] Placing a water-filled balloon between the transducer head and the skin, with coupling gel at the interfaces, has been described.[10] A recent in vitro study compared the transmission properties of commercial gel pads, bladder techniques, and water bath immersion, through pig skin from which the hair had been clipped.[14] Insufficient acoustic energy transmission occurred with degassed tap water in latex gloves, gel in a latex glove, degassed tap water baths, and gel-filled condoms. The authors recommended that if the direct method cannot be used because of the contour of the surface to be treated, clinicians should consider the use of commercially available gel pads. If other direct techniques are used, transmission is considerably less than with the direct coupling technique or commercial gel pads.[14]

The Effect of the Hair Coat

The hair coat of dogs treated with US presents a problem not encountered with human patients. Because US energy is absorbed by tissues with high protein content, and deflection of the US beam occurs at tissue interfaces, it would be expected that US penetration through the hair coat into underlying tissues would be poor. For horses, one author has specifically recommended clipping the hair and using adequate coupling gel or having the horse stand in water to reduce any air interphase.[17]

A study was conducted in dogs to determine temperatures in underlying tissues when delivering US through intact hair coats.[18] Despite application of a thick layer of gel on the skin, US delivered through short or long hair coats produced only minimal temperature increases in the underlying tissues, compared with therapy after the hair had been clipped, although there was considerable warming within the hair coat and on the skin (Figure 19-1).

Treatment Variables

The following list contains treatment variables that must be considered for each patient. These items should be documented in the patient's record:

1. Frequency
2. Intensity
3. Duty cycle
4. Treatment area
5. Treatment duration
6. Speed of the sound head
7. Treatment schedule

Sound waves above a frequency of 20,000 hertz (20 kHz) exceed the upper range of human hearing and are termed US. The higher the frequency, the less divergent the sound beam. Ultrasound beams are considered to be collimated, and frequencies in the megahertz (MHz) range expose a limited target area.

Frequency, not intensity, determines depth of penetration. The frequency most often used is 1 MHz because it represents a compromise between deep penetration and adequate heating. A frequency of 1 MHz frequency heats at depths between 2 and 5 cm.[19] As the frequency increases, penetration decreases. A frequency of 3.3 MHz (commonly referred to as 3 MHz) heats at depths between 0.5 and 3 cm.[8,19,20] Therefore for superficial lesions, a frequency of 3 MHz is preferred; for deeper lesions, a setting

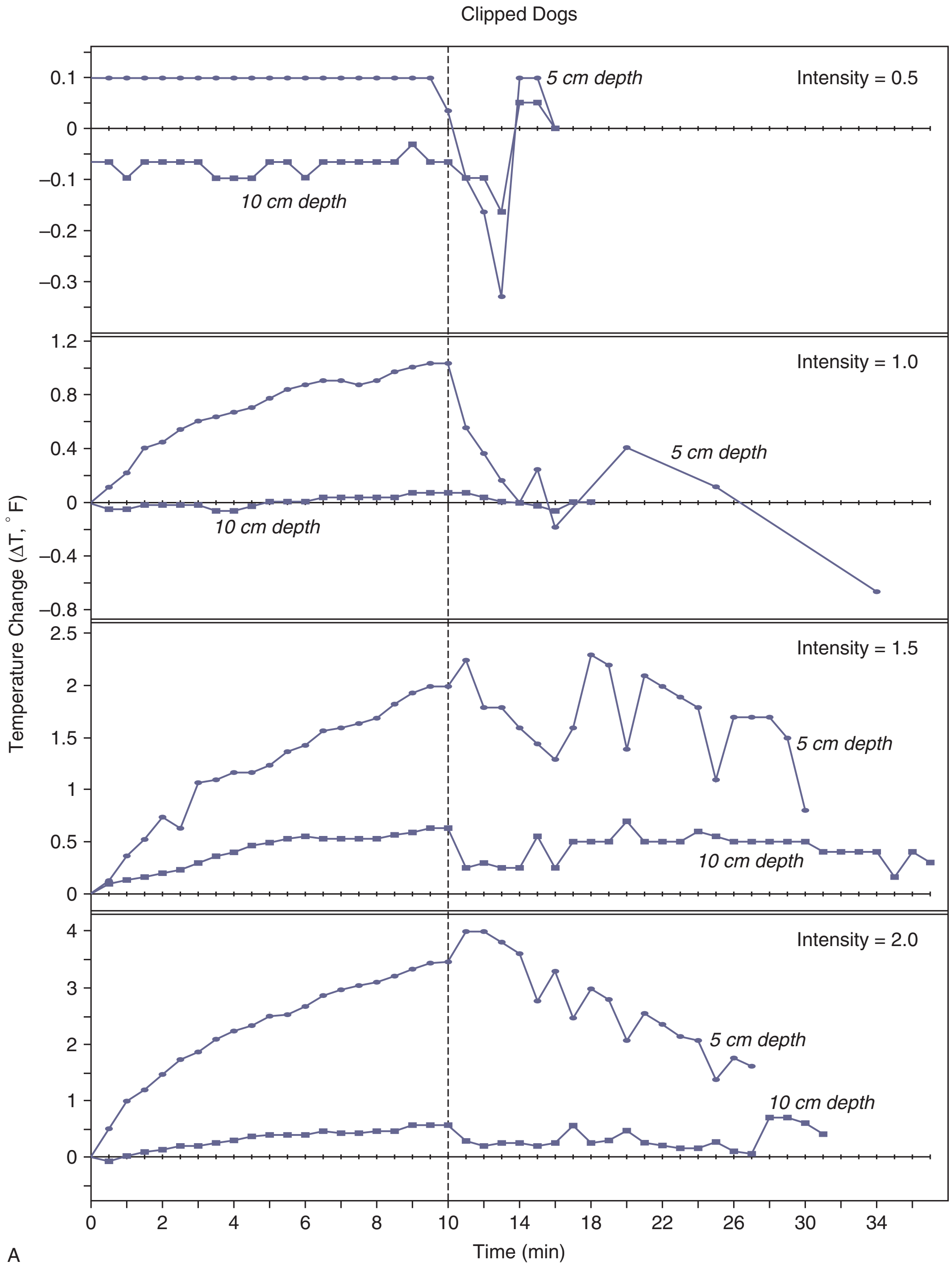

Figure 19-1 **A,** Intramuscular temperature changes measured at 5- and 10-cm depths during 1-MHz ultrasound at four intensities (W/cm^2) applied to clipped skin over the caudal thigh in dogs. The vertical dashed line represents the end of 10 minutes of ultrasound exposure. The units on the y-axis differ for each intensity.

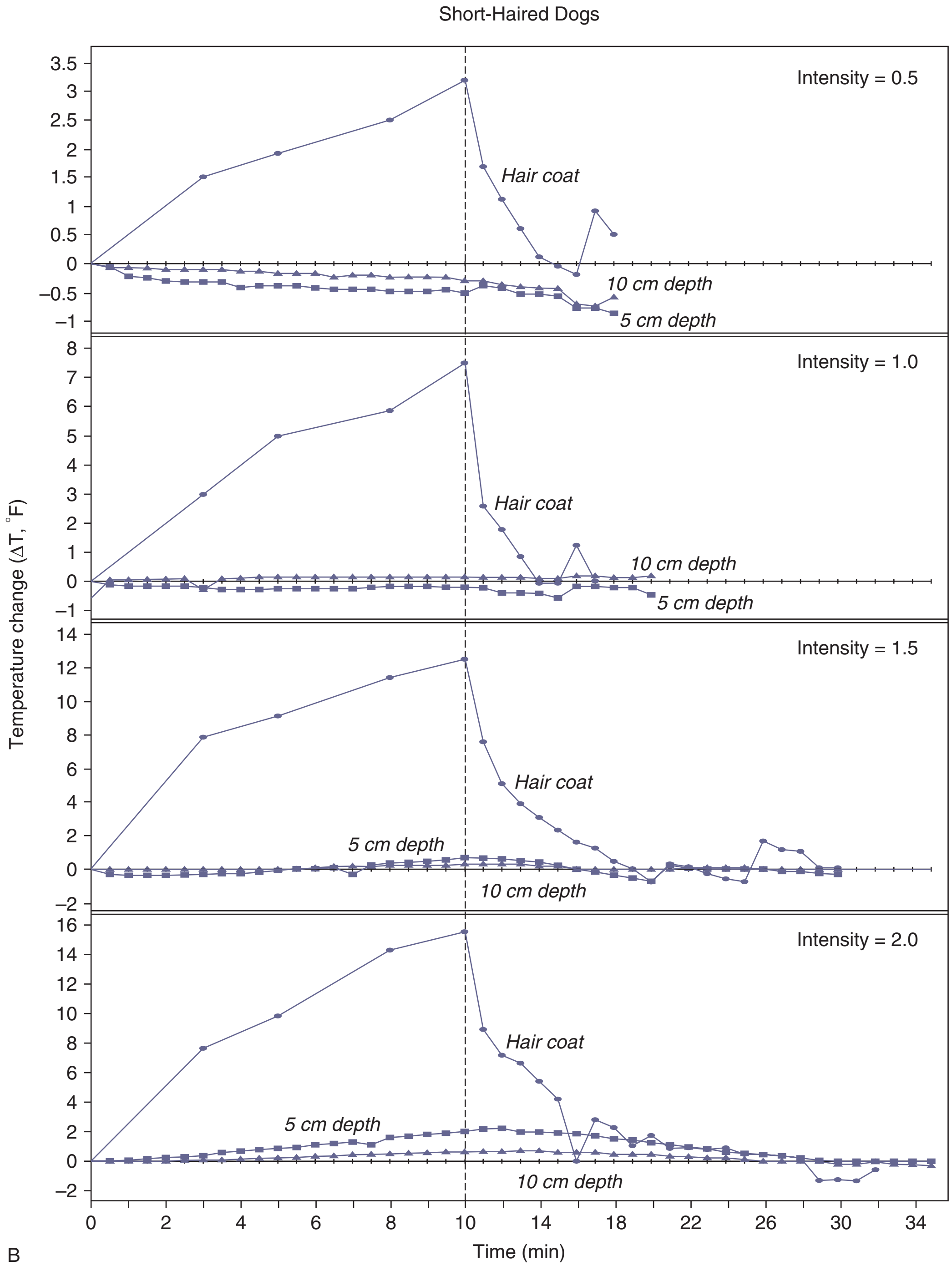

Figure 19-1, cont'd **B,** Temperature changes measured intramuscularly at 5- and 10-cm depths and within the hair coat, during 1-MHz ultrasound at four intensities (W/cm^2) applied over the caudal thigh in dogs with short hair coats (greyhounds). The vertical dashed line represents the end of 10 minutes of ultrasound exposure. The units on the y-axis differ for each intensity. (From Steiss JE, Adams CC: Effect of coat on rate of temperature increase in muscle during ultrasound treatment of dogs. *Am J Vet Res* 60:76-80, 1999.)

of 1 MHz is needed.[8] If there is underlying bone, most of the US transmitted to that depth will be absorbed by the bone and may cause periosteal pain.[19-21] The intensity should not be turned up very high to heat the underlying soft tissue if the bone is not adequately covered by soft tissues.[19] Low frequency or "long wave" US machines have also been investigated.[22] These units emit US at a lower frequency, around 45 kHz. In an in vitro study using fibroblasts, osteoblasts, and monocytes, both 45 kHz and 1 MHz exposure enhanced cell proliferation, protein synthesis, and cytokine production.[22]

Intensity refers to the rate of energy delivery per unit area. Most US machines indicate both watts (W) and watts per centimeter squared (W/cm^2). Available intensities typically range from 0.25 to 3.0 W/cm^2. Measurement of the total output (W) of the transducer divided by the transducer surface area (cm^2) is referred to as the spatial average intensity. The greatest intensity anywhere within the beam is termed the spatial peak intensity. The beam nonuniformity ratio (BNR) compares the maximal intensity from the transducer to the average intensity. This ratio should be low (between 2 and 6) to indicate that the energy distribution is relatively uniform, with minimal risk of tissue damage from areas of concentrated US energy ("hot spots"). This risk is minimized when the therapist keeps the US transducer moving during treatment.

The higher the intensity, the greater and faster the temperature increases. The depth of tissue penetration is not affected to a large degree. Generally intensities required to increase tissue temperature 2° C or more vary from 1 to 2 W/cm^2 continuous wave US for 5 to 10 minutes. To heat an area with substantial soft tissue, intensities as high as 2 W/cm^2 may be used. If there is less soft tissue, or if bone is close to the skin surface, lower intensity and higher frequency are appropriate. After selection of an initial intensity, the patient's tolerance to the heat produced by the US is the final determinant of intensity.[7] This may be difficult to determine in pets. Many dogs lie quietly during US treatment. However, dogs sometimes begin to whine or otherwise seem uncomfortable within 5 to 10 minutes after commencing US. It should be assumed that any distress demonstrated by a patient may indicate pain, and the intensity of the US should be reduced or the session interrupted or discontinued.

Most therapists tend to use intensities that produce no detectable sensation in human patients. However, some clinicians advise that the intensity be increased until the person feels vigorous heating, after which time the intensity should be decreased if the patient experiences pain.[19] This is impractical in dogs. The US dosage that a patient is receiving usually cannot be monitored. In the past, in an effort to avoid tissue damage, the dosages have been low and the patient experiences no sensation. However, there is the possibility that the patient is not receiving a sufficient dose of US energy. More recent recommendations seem to address this issue.

When the treatment goal is to heat periarticular structures, as is done when treating limited ROM, higher intensities are chosen. Lower intensities may be used for the relief of pain and muscle spasm, in which the mechanism of action may be nonthermal.

Duty cycle refers to the fraction of time that the US is emitted during one pulse period. With continuous mode the US intensity is constant, whereas a pulsed wave is interrupted so that energy is delivered in an on/off manner.

$$\text{Duty cycle} = \frac{\text{Pulse duration (time on)}}{\text{Pulse period (time on + time off)}}$$

Typical duty cycles for pulsed mode US range from 0.05 (5%) to 0.5 (50%). For pulsed mode, the terms temporal peak intensity and pulse average intensity are used. Obviously pulsing decreases the pulse average intensity and decreases tissue heating. Pulsing may be used when the desired effect is based on a nonthermal mechanism or when minimal heating is desired, such as treating near bone.[8] Low-intensity continuous mode could also be chosen for these purposes.

The treatment area should be two to four times the size of the effective radiating area of the transducer head.[19] Increasing the total area beyond the recommended area decreases the dosage and the heating effect.

Experimental evidence indicates that a duration of 5 to 10 minutes is necessary to produce adequate tissue heating in an area equivalent to two to three times the diameter of the sound head.[21,23] For instance when using direct coupling with topical US gel, continuous mode at 1.5 W/cm^2 for 10 minutes, and moving the sound head at 4 cm per second, it took nearly 8 minutes for the temperature to reach therapeutic levels when temperature was measured at a depth of 3 cm in the gastrocnemius muscle of humans.[12] For an intensity of 1.5 W/cm^2 Draper recommends a duration of approximately 5 minutes at 3 MHz and at least 10 minutes at 1 MHz, assuming that an area twice the size of the effective radiating area of the transducer is treated.[19] The findings in dogs (see Figure 19-1, *A*) support these recommendations.

The recommended speed at which the sound head is moved over the skin is approximately 4 cm per second, or slower, to achieve uniform distribution of energy to the target tissues.[10] Moving the transducer too quickly diminishes the heating and the therapist may have a tendency to cover too large an area. Various longitudinal and circular patterns may be used. The transducer should never be stationary when direct coupling is used. Because the US beam is nonuniform, some target areas within the beam could receive a large amount of energy. This predisposes target areas to hot spots and tissue damage, including endothelial damage and platelet aggregation. Keeping the transducer moving is not as critical with underwater applications because hot spots occur in the near field of the beam. Hot spots are situated in the water if a sufficient distance is maintained from the tissue. However, the distance may compromise the heating effect. Problems with achieving adequate tissue heating with the immersion technique have been previously described.

Treatment schedules may include daily treatment initially, followed by less frequent sessions as the condition improves.[7] Bromiley[8] states that treatment can be given daily for up to 10 days, but should not exceed two 10-day courses without a 3-week rest. Similar recommendations have been made for horses.[17] However, the scientific rationale for these recommendations is not clear.

Precautions and Contraindications

Tissue burns are a major concern of treatment.[13] These may occur if the intensity is too high, treatment times are too long, or the transducer is held stationary, thereby concentrating energy in a small area. These same factors may put the patient at risk for cavitation, a phenomenon in which bubbles of dissolved gas form and grow during each rarefaction phase.[10] Also if the transducer is inadvertently held in the air while emitting US, the face of the transducer may overheat. The transducer may become damaged, and the animal could be burned if the overheated surface of the transducer contacts the skin. Some units have a built-in system to prevent the crystal from overheating. Regarding safety of the therapist, some US energy is unavoidably propagated through the housing of the sound head to the hand of the therapist, but the effects of this "parasitic" exposure are unknown.

Avoid direct US exposure to the following:

- Cardiac pacemakers.
- Carotid sinus or cervical ganglia. Ultrasound could alter the normal pacing of the heart or stimulate baroreceptors.[7]
- Eyes. Because blood supply to the lens is poor, heat is poorly dissipated and could result in cataracts. Retinal damage could also result.[10]
- Gravid uterus.
- Heart. Electrocardiographic changes have occurred in dogs.
- Injured areas immediately after exercise.[15,17]
- Malignancy. Ultrasound applied to murine tumors resulted in larger tumors compared with controls.[24]
- Spinal cord if a laminectomy has been performed.[7] However, US therapy has been advocated in the area around the spinous processes in horses[25] and would also be acceptable in dogs if the bone is intact.
- Testes. Temporary sterility could be associated with heating.
- Wounds that are contaminated. Ultrasound could drive bacteria into the tissues[26]; infectious processes may be accelerated by heat.[7]
- Recent incision sites. It has been recommended not to treat incision sites with continuous US for the first 14 days to avoid dehiscence.[17]
- **Exert precaution in the following situations:**
- Bone fracture. This is controversial. Concern exists that high doses retard callus formation, delay calcification,[15] or cause pathologic fractures, subperiosteal damage, or demineralization.[26] Others believe that US over fracture sites is acceptable unless sensation is impaired.[9] Experimentally, low-intensity pulsed US may accelerate fracture healing.[10]
- Bony prominences. Treat around bony prominences rather than directly over them, to avoid concentrating energy at the periosteum, or use an immersion technique. Treat bony areas with low dosage levels to avoid the danger of pathologic fractures.[8]
- Cold packs or ice before US. Cold alters pain and temperature perception, limiting the patient's ability to respond if the US intensity is too high.
- Decreased blood circulation. Normal blood flow dissipates the heat that is generated.
- Decreased pain and temperature sensation. Pain response serves as a sensitive monitor of excessively high intensities or faulty equipment[17] that could cause burns or other tissue damage.
- Animals that are overly sedated, restrained, or under local anesthesia.
- Physeal areas in immature animals.[8,10,15] High intensities could affect physes and bone growth.
- Injuries in the acute stage that should not receive heat therapy.[17]
- Acute inflammatory joint disease. Intracapsular heating may accelerate destruction of articular cartilage in acute inflammatory joint disease.[7]

Note that metal implants are not necessarily a contraindication for US therapy. The effects of US on cementing compounds such as methylmethacrylate are unknown.[10]

Equipment Safety

Ultrasound equipment falls under the rules and regulations of the FDA's radiation safety performance standards.[15] Verification of such features as output accuracy and timer accuracy, as well as safety relating to the electrical components, is advised semiannually.[10] Ultrasound power meters are available for testing. The manufacturer establishes the BNR of a transducer. Most equipment problems originate in the transducer or cable assembly. Recalibration is necessary when a transducer is replaced, and yearly calibration by the manufacturer is recommended even if the transducer is not replaced. Some machines feature self-calibration tests, output power control in response to tissue loading, and automatic shutoff in case of transducer overheating.

Thermal Effects

Early applications of therapeutic US were those for which tissue heating was the goal.[26] More recently, attention also has been drawn to low-intensity US, which appears to stimulate physiologic processes, such as tissue repair, without significant tissue heating.[27]

The thermal effect is a major indication for the use of this modality. Increased tissue temperatures may increase collagen extensibility, blood flow, pain threshold, and enzyme activity, as well as mild inflammatory reactions, and changes in nerve conduction velocity. Heating is thought to alter the viscoelastic properties of collagen and collagen molecular bonding. Changes in blood flow in response to US have been measured with variable results.[28,29] It is likely that with treatment for 10 to 20 minutes at high intensities, skeletal muscle temperature and blood flow increase.[10] Inconsistent changes have been found with lesser intensities and durations.

The degree of heat production depends on the intensity, frequency, duration, and size of the treatment area. In addition, the tissue type must be considered. Ultrasound penetrates skin and subcutaneous fat relatively well, with little attenuation, whereas tissues with high collagen content absorb more of the US energy and are affected to a greater extent. The order of tissue attenuation, from least to most, is as follows: blood (approximately 3% attenuation), fat, muscle, blood vessels, skin, tendon, cartilage, and bone (approximately 96% attenuation).[10]

To achieve the thermal effects of US, the tissue temperature must be raised 1° to 4° C, depending on the desired outcome. Obviously this is difficult to judge because thermistors are not inserted in clinical patients to monitor the temperature. Experimental studies in dogs and humans can provide guidelines. In healthy adult people thermistors were inserted into muscle at depths of 2.5 and 5.0 cm for 1 MHz treatment, and at depths of 0.8 and 1.6 cm for 3 MHz treatment.[30] The rate of temperature increase per minute at the two depths for the 1 MHz frequency ranged from 0.04° C at 0.5 W/cm^2 to 0.38° C at 2.0 W/cm^2, whereas the corresponding values for the 3 MHz frequency ranged from 0.3° C at 0.5 W/cm^2 to 1.4° C at 2.0 W/cm^2. The 3 MHz frequency heated significantly faster than 1 MHz at all doses.

One method for treating humans consists of applying ice on the treatment area before US. Adding ice to the water for US delivered underwater has been recommended for horses also.[25] However, if the aim is to increase tissue temperature, then it appears that US preceded by ice treatment yields little benefit. Experimentally, US alone increased tissue temperature an average of 4° C, whereas US preceded by 5 minutes of ice increased tissue temperature only 1.8° C.[2]

One prospective randomized experimental study examined the tissue temperature changes that occur at various depths during 3.3 MHz US treatments of the caudal thigh muscles in dogs.[20] Dogs received two US treatments (selected randomly), one at an intensity of 1.0 W/cm^2 and one at 1.5 W/cm^2. Needle thermistors were inserted in the caudal thigh muscles below the skin surface at depths of 1, 2, and 3 cm, directly under the US treatment area. Both intensities of US treatment were performed on each dog over a 10-cm^2 area for 10 minutes using a sound head with an effective radiating area of 5 cm^2. Tissue temperature was measured before, during, and after US treatment until tissue temperature returned to baseline. At the completion of the 10-minute US treatment the temperature rise at an intensity of 1.0 W/cm^2 was 3° C at the 1-cm depth, 2.3° C at 2 cm, and 1.6° C at 3.0 cm (Figure 19-2, *A*). At an intensity of 1.5 W/cm^2 tissue temperatures rose 4.6° C at the 1.0-cm depth, 3.6° C at 2 cm, and 2.4° C at 3 cm (Figure 19-2, *B*). Tissue temperatures returned to baseline within 10 minutes after treatment in all dogs. This study demonstrated that significant heating occurs in the superficial thigh muscles of dogs during 3.3 MHz US, but the effect is relatively short-lived.

Increasing tendon extensibility with the use of US was also studied in dogs.[31] This study evaluated tendon temperature of the calcaneal tendon following US treatment, and quantified hock flexion following US treatment of the tendon. Dogs were anesthetized and the calcaneal tendon region was clipped free of hair. One leg was used to measure temperature increase of the tendon with four modes of US: continuous and pulsed mode US, at both 1.5 W/cm^2 and 1.0 W/cm^2. The other leg was used to measure the change in hock flexion using a standard amount of force, pre- and post-US treatment of the calcaneal tendon using 1.5 W/cm^2 continuous-mode US. Continuous-mode US resulted in significantly greater tendon heating than pulsed US ($P < .001$). In addition, 1.5 W/cm^2 continuous US resulted in significantly greater tendon heating than 1 W/cm^2 continuous US ($P < .001$). Mild increases in tendon temperature occurred with pulsed mode US (all < 1.5° C), but there were no differences between 1 and 1.5 W/cm^2 pulsed mode US. The pattern of heating appears to be different between tendon and muscle. Muscle appears to have a relatively steady increase in tissue temperature during US treatment, whereas tendon temperature in this study increased relatively rapidly within the first 2 minutes, and then stayed relatively constant for the remaining 8 minutes of treatment time. Continuous mode 1.5 W/cm^2 US

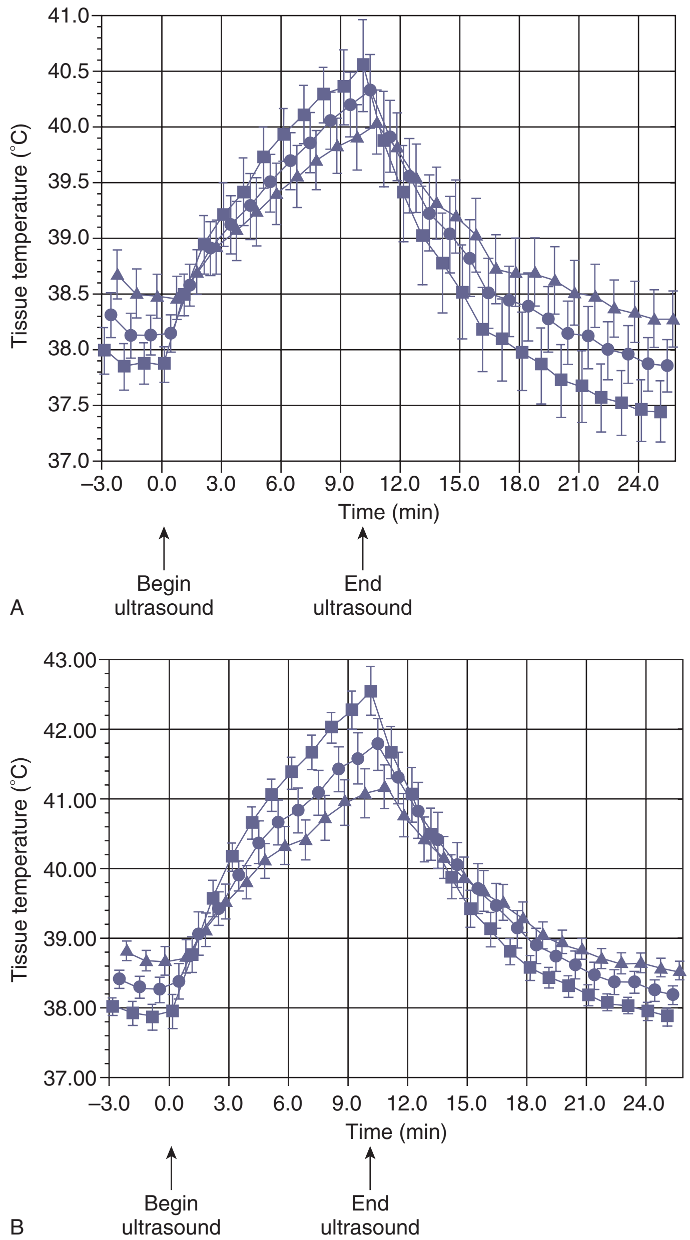

Figure 19-2 **A,** Baseline, US treatment, and posttreatment tissue temperatures for 1.0 W/cm^2, 3.3-MHz ultrasound at 1-, 2-, and 3-cm tissue depths. Values are means ± SEM. **B,** Baseline, US treatment, and posttreatment tissue temperatures for 1.5 W/cm^2, 3.3-MHz ultrasound at 1-, 2-, and 3-cm tissue depths. Values are means ± SEM. (From Levine D, Millis DL, Mynatt T: Effects of 3.3 MHz ultrasound on caudal thigh muscle temperature in dogs. *Vet Surg* 30:170, 2001.)

treatment of the calcaneal tendon resulted in significantly increased hock flexion immediately after treatment, but by 5 minutes post-US, hock flexion returned to near baseline.

Clinical Conditions Treated with Ultrasound

The main indications for US therapy include the following:

- Soft tissue shortening (e.g., contracture, scarring)
- Subacute and chronic inflammation
- Pain (such as muscle guarding, trigger points)

In human athletes, some experts consider that the most beneficial results from US are in treating tendinitis.[19] In chronic tendinitis, recommended therapy includes heating with US, followed by cross-frictional massage, to eliminate scar tissue and increase ROM. Another indication is the treatment of limited ROM associated with joint contracture, for which patients receive US before passive

ROM, stretching, or joint mobilization. A third indication is for pain relief before activity, such as in athletes with tendinitis that is mild enough to allow the person to continue training. The US treatment is administered before activity to assist in the warm-up and provide some pain relief.

In horses, US therapy has been suggested for the treatment of similar conditions. These include tendinitis, desmitis, sprains, joint lesions, lacerations, reduction of scar tissue, edema, exostosis, and myositis.[17]

Tendinitis and Bursitis

Tendinitis and bursitis are treated with US because of the ability to increase blood flow to aid healing, increase tissue temperature to reduce pain, and drive antiinflammatory drugs across the skin by means of phonophoresis. In humans lateral epicondylitis (tennis elbow), subacromial bursitis, and bicipital tendinitis are typical conditions considered for US therapy. From animal studies it appears that the stage at which US is given in the course of healing is important. Application of US in the early phase of tendon repair could be detrimental.[10]

Joint Contracture and Scar Tissue

Ultrasound is often included in protocols for treating joint contractures and scar tissue.[10] Joint contracture typically results from immobilization or trauma. The periarticular connective tissues have high collagen content, and if tissues are not mobilized, they tend to shorten to their resting, immobilized length. The principle of "heat and stretch" can be applied in this situation in an effort to increase ROM. Tissues are heated to approximately 45° C, followed by gentle passive or active stretching. There tends to be a greater residual increase in tissue length with either preheating or simultaneous heating. The effects of heat on ligament extensibility have been studied in humans.[32] Young adults underwent knee joint displacement tests before and after continuous-mode US (1 MHz, 1.5 W/cm^2 for 8 minutes). The investigators found small increases (1.3 degrees) in varus/valgus excursion at 0- and 20-degree knee flexion and in recurvatum excursion, but not in anterior-posterior drawer excursion. Their conclusion was that continuous-mode US at commonly used intensities made some knee ligaments only slightly more extensible in normal subjects. Scar tissue is denser than surrounding tissues, so it may be selectively heated by US. However, more research is needed to determine which intensities and durations are most beneficial. Figure 19-3 demonstrates US treatment over the hip before stretching in a patient who has undergone a femoral head and neck ostectomy.

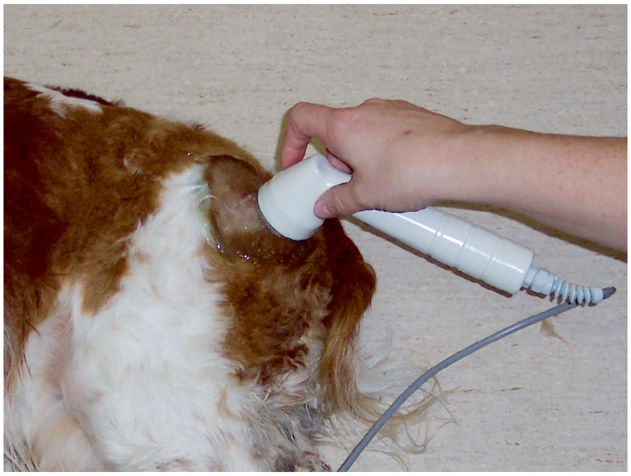

Figure 19-3 Ultrasound applied over the hip joint before stretching in a patient with a femoral head ostectomy.

Pain and Muscle Spasm

Pain threshold is usually increased after US. Diathermy and infrared therapy have a similar effect.[10] Although the physiologic mechanism underlying pain reduction remains speculative, the probable mechanism of action is the ability of all three methods to elevate tissue temperature. Heating may elevate the activation threshold of free nerve endings, produce counter-irritation, or activate large-diameter nerve fibers.[10] Peripheral nerve conduction velocities are either increased or decreased after US, partially depending on the intensity. Types A, B, and C nerve fibers have different sensitivities.[8] With intensities of 1 to 2 W/cm^2 US reduces the nerve conduction velocity of pain-carrying C fibers. This may help to reduce muscle spasm and reestablish circulation. Ultrasound also has been used to treat pain associated with neuromas, low-back dysfunction, and skeletal muscle spasm. The mechanism of action for reduction of skeletal muscle spasm may be based on thermal effects that alter the skeletal muscle contractile process, reduce muscle spindle activity, or break the pain-spasm-pain cycle.[10]

Nonthermal Effects of Ultrasound to Enhance Tissue Repair

There are many situations in which US produces biologic effects and yet significant temperature change is not involved. It is not strictly true to talk about "nonthermal" mechanisms, in that the delivery and absorption of energy in the tissues will result in a small temperature rise. The term *nonthermal* in this context relates to the fact that there is no apparent thermal accumulation in the tissues, and is sometimes now referred to as a *microthermal* effect.[33] The physical mechanisms thought to be involved in producing these nonthermal effects include cavitation and acoustic streaming.[27,34]

There is evidence indicating that nonthermal mechanisms play a primary role in producing a therapeutically significant effect, and the focus of this section relates to these effects in relation to soft tissue injuries because these are the most commonly treated conditions in practice.

Ultrasound in Soft Tissue Repair

The process of tissue repair involves a complex series of cascaded, cellular and chemically mediated events that lead to the production of scar tissue that constitutes an effective material to restore the continuity of the damaged tissue. The process is more complex than can be described in this chapter, but has been extensively reviewed elsewhere.[35]

The effect of US during the repair process varies according to the primary events that are occurring in the tissues. Immediately following injury, during the tissue hemorrhage phase, it is generally inadvisable to use US because if it has the capacity to enhance local blood flow, and there may be a clinical disadvantage of treatment while hemorrhage is occurring. After active bleeding has stopped (usually a matter of hours after injury or trauma), it is appropriate to start using US as soon as it is feasible. During the inflammatory phase, US has a stimulating effect on the mast cells, platelets, neutrophils, and macrophages.[27,36-38] For example, the application of US induces the degranulation of mast cells, causing the release of arachidonic acid, which is a precursor for the synthesis of prostaglandins and leukotrienes that act as inflammatory mediators[36,39-41] By increasing the activity of these cells the overall influence of therapeutic US is proinflammatory rather than antiinflammatory. The benefit of this mode of action is not to "increase" the inflammatory response as such (although if applied with too great an intensity at this stage, it is a possible outcome),[41] but rather to act as an "inflammatory optimizer" by speeding the process up. The inflammatory response is essential to the effective repair of tissue, and inhibition of these events serves to inhibit the repair phases that follow.[35,42-44] The more efficiently the inflammatory events can be completed, the more effectively the tissue can progress to the proliferative phase of repair. Studies that have tried to demonstrate the antiinflammatory effect of US have largely failed to do so.[45,46] It is effective at promoting the normal inflammatory events, and as such has a therapeutic value in promoting the overall repair events.[27,33,47,48]

Employed at an appropriate treatment dose with optimal treatment parameters, the benefit of US is to make the inflammatory phase as efficient as possible, and thus promote the entire healing cascade.[7,10,35,49] For tissues in which there is an inflammatory reaction, but in which there is no "repair" to be achieved, the benefit of US is to promote the normal resolution of the inflammatory events, and hence resolve the "problem." This is most effectively achieved in the tissues that preferentially absorb the energy, which are the dense collagenous tissues (ligament, tendon, fascia, joint capsule, and scar tissue). During the proliferative phase of wound healing US also has a stimulatory effect (cellular upregulation), although the primary active targets are the fibroblasts, endothelial cells, and myofibroblasts.[36,38,39,50-54] These are all cells that are normally active during scar production and US is therefore pro-proliferative in the same way that it is proinflammatory—it does not change the normal proliferative phase, but maximizes its efficiency. Harvey et al.[55] demonstrated that low-dose pulsed US increased protein synthesis, and several research groups have demonstrated enhanced fibroplasia and collagen synthesis.[50,56-60]

During the remodeling phase of repair, the somewhat generic scar that is produced in the initial stages is refined such that it adopts functional characteristics of the tissue that it is repairing.[35] This is achieved by a number of processes, but is mainly related to the reorientation of the collagen fibers in the developing scar and also to the change in collagen type, from predominantly Type III collagen to a more dominant Type I collagen.[61] The remodeling process is certainly not a short-duration phase (it can last for a year or more), yet it is an essential component of quality repair and the application of US appears to enhance the events normally associated with this phase.

Ultrasound can influence the remodeling of scar tissue in that it appears to enhance the appropriate orientation of the newly formed collagen fibers[62] and change the collagen profile from mainly type III to a more dominant type I content, thus increasing tensile strength and enhancing scar mobility.[53] Other studies have demonstrated effects of US therapy on collagen behavior in this latter repair stage.[63-66] Ultrasound applied to scar tissue may enhance its function.[53,60]

The application of US during the inflammatory, proliferative, and repair phases is of value not because it changes the normal sequence of events, but because it has the capacity to stimulate or enhance these normal events and thus increase the efficiency of the repair phases.[27,33,47] It appears that if tissue repair occurs in a compromised or inhibited fashion, the application of therapeutic US at an appropriate dose may enhance healing. If the tissue is healing "normally," the application may speed the process and thus enable the tissue to reach its end point faster than it would otherwise. The effective application of US to achieve these aims may be dose dependent,[49,64,67] although some authors have challenged this thought.[68]

Soft Tissue Wound and Fracture Healing

The use of therapeutic US in wound healing recently has been reviewed (Figure 19-4).[69] Reher et al.[70] suggested that the beneficial effects of US on wound healing, chronic ulcers, and fracture healing might be based on enhanced angiogenesis. Therapeutic US stimulates production of angiogenic factors such as interleukin-8, fibroblast growth factor, and vascular endothelial growth factor. These effects were noted with both 1-MHz and 45-kHz exposure. More research is needed to establish the mechanism of action of US at different stages of healing and with different treatment parameters to optimize its effects.

Results may depend on variables such as the intensity and duration of treatment and the timing of application after injury. Low-intensity continuous- or pulsed-mode US has been used to treat acute and chronic wounds. Low intensities appear to enhance healing, whereas high intensities may have proinflammatory effects. Regarding the stage of healing, US therapy initiated while hemorrhage is still occurring may compromise repair, whereas US initiated after this may be beneficial.

Dyson et al.[71] reported that US enhanced growth of tissue in experimental wounds in rabbit ears. Because the temperature increases were small, the investigators suggested that the mechanism may have involved acoustic streaming. Ultrasound has also been used to treat chronic skin ulcers. Rantanen et al.[72] used a rat model of muscle contusion injury to study the effect of therapeutic US on muscle regeneration. They concluded that therapeutic US promoted the early phase of muscle regeneration, namely, the proliferation of satellite cells, but US did not seem to significantly alter the final outcome. Low-intensity US may also have a role in the healing of fractures. Numerous animal studies and controlled clinical trials have shown that low-intensity US can promote the healing of fresh fractures. Some evidence suggests efficacy in the treatment of delayed healing and nonunion fractures as well.[73-75]

Other Conditions

There was significant improvement after US treatment in people with mild to moderate idiopathic carpal tunnel syndrome.[76] Patients received 20 US sessions (1 MHz, 1 W/cm^2, pulsed mode 1 : 4, 15 minutes per session) applied to the area over the carpal tunnel, with the first 10 treatments performed daily and 10 additional treatments performed twice weekly for 5 weeks. The improvements were sustained 6 months after treatment. It remains to be determined if any forms of nerve compression or entrapment in dogs would benefit from a similar protocol.

Figure 19-4 Ultrasound applied to a wound on the lateral aspect of the stifle joint through a stand-off gel pad.

Ultrasound treatment may be useful in the treatment of soft tissue calcification. In a randomized, double-blind comparison of US- and sham-treated human patients with symptomatic calcific tendinitis of the shoulder, US had a positive effect in resolving calcifications and was associated with short-term clinical improvement.[77] Patients in that study received 24 sessions of pulsed US over a 6-week period. The way in which US stimulates resorption of calcium deposits has not been established. Bromiley[8] has suggested that the excitation of calcium bound to proteins promotes the fragmentation and resorption of calcified masses within soft tissue. Ebenbichler et al.[77] speculated that US may stimulate resorption of calcium deposits by (1) activating endothelial cells that subsequently release substances such as cytokines and interleukins that aid in resorption, (2) increasing intracellular calcium levels, (3) at high intensities, causing disruption of apatite-like microcrystals, and (4) causing increased temperatures that may lead to increased blood flow and metabolism, thereby facilitating disintegration of calcium deposits.

Ultrasound may also have a role in reduction of swelling. The mechanism has been attributed to acoustic streaming.[26] For this purpose pulsed US may be combined with cold therapy.

Dosage

Dosage for pulsed US is variable based on the tissue and the stage of healing. In general, 20% pulsed US with an intensity of between 0.1-1.0 W/cm^2 is used.[35,37]

Phonophoresis

The use of US to enhance the delivery of topically applied medications through intact skin is termed *phonophoresis*.[78] The target for the drug may be local (antiinflammatory or analgesic) or systemic. Some of the local conditions for which phonophoresis is used in humans and that parallel conditions in animals are tendinitis, bursitis, chronic sprains, rheumatoid arthritis, and osteoarthritis (OA).[79] Systemic conditions that may be treated by phonophoresis include motion sickness and high blood pressure. Although many studies have been conducted on humans and research animals to evaluate the penetration of various drugs through the skin with the assistance of US, few studies have been performed on companion animals with clinical conditions. Therefore more research is needed to better define the benefits and limitations of this rehabilitation modality.

When a drug is applied topically it may transfer through the skin by passive diffusion.[80,81] This may be hastened by denuding the epidermis to create minor abrasions, such as in shaving. Phonophoresis enhances transdermal drug delivery by altering the stratum corneum, which is the major barrier to drug penetration.[82,83] The mechanism of action may be that US denatures the structural keratin proteins in the stratum corneum, strips or delaminates the cornified layers of the stratum corneum, changes cell permeability, or alters the lipid-enriched intercellular structures between the cells.[78] After the stratum corneum is crossed the pharmaceutical molecules penetrate the dermis, are absorbed into the tissues and capillaries, and travel throughout the circulatory system.[80] In animals, shaving and scrubbing the skin with alcohol increase the rate of diffusion.[78] Drug penetration may occur transcellularly or intercellularly, but it occurs most readily through the hair follicles and sebaceous glands.[84] Although heating the skin before treatment dilates the hair follicles to enhance penetration, it also causes vasodilatation that enhances systemic absorption and decreases local delivery. Low-intensity, longer duration US appears to be more effective at transferring medication than high-intensity, shorter duration doses.[85] Higher concentrations of drugs applied topically also may increase tissue concentrations.[79,86]

Another variable to consider is the transmission of US energy through the drug and coupling agent. Some compounds (with such active ingredients as diethylamine salicylate, hydrocortisone, alclometasone dipropionate, and fluclorolone acetonide) are used to decrease soft tissue inflammation and did not allow transmission of US at three different frequencies (0.75, 1.5, and 3 MHz) in one study.[87] Because the sound waves were unable to penetrate the drug, they were also unable to penetrate the skin. Not only could using a drug that has poor transmission properties decrease the success of phonophoresis, it may also negate the effectiveness of the US therapy. Intensity, frequency, time, drug, and coupling agent transmission may play a role in some studies of phonophoresis that were unsuccessful.[88,89]

Several studies have been performed using steroidal antiinflammatory drugs (hydrocortisone and dexamethasone) and have shown decreased pain and increased ROM in human patients with OA, periarticular arthritis, and joint or muscle pathology.[90] In a human study using hydrocortisone phonophoresis (100 mg/g of ointment) on patients with OA, tendinitis, or bursitis, there was improvement (marked decrease in pain and a substantial gain in active ROM) in 68% of patients and partial improvement (decrease in pain but some pain remaining, and a gain in active ROM with limitations still present) in 18% of patients as compared with 27% improvement and 16% partial improvement when treated with US alone.[90] In one study using dexamethasone sodium phosphate (7 g of 0.33% dexamethasone using 1 MHz at 1.5 W/cm^2) in humans, the amount of dexamethasone delivered systemically did not alter adrenal function.[91] Another study of phonophoresis in humans showed detectable blood levels of dexamethasone sodium phosphate 30 and 60 minutes after treatment, whereas none was detected in the blood of patients that had the drug applied topically with no US.[92]

When using phonophoresis on small animals, the skin should be clipped and scrubbed. The drug may be either applied topically with a gel over it or compounded in glycerol, cream, oil, gel, or water. Many researchers believe that gels are optimal because of their ability to transmit US.[92] Pulsed US is most commonly used because continuous US increases blood flow to the area and may exacerbate inflammatory conditions. One recent human study compared ketoprofen phonophoresis (both 20% pulsed and 100% or continuous US) to placebo phonophoresis in which ketoprofen gel was rubbed over the area. Biopsies were performed on both adipose and synovial tissues of the knee approximately 1 hour after phonophoresis application. No significant differences were found between groups in the adipose tissue samples. In synovial tissues, both pulsed and continuous US produced significantly higher concentrations of ketoprofen compared with topical application ($P < .012$ and $P < .028$, respectively).[93]

One of the essential benefits of phonophoresis is that it may deliver higher drug levels to the target area, decreasing systemic side effects as compared with parenteral or oral medications.[94] Additionally, US is relaxing and may have analgesic properties.[4-6]

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